

Chapter 20

Optimization Under Uncertainty

Algorithms for Optimization, 2nd Edition (2026)

Kochenderfer & Wheeler

Slides prepared for lecture

Chapter Overview

- 1 Types of Uncertainty
- 2 Set-Based Uncertainty (Minimax)
- 3 Probabilistic Uncertainty
- 4 Summary & Exercises

Outline

- 1 Types of Uncertainty
- 2 Set-Based Uncertainty (Minimax)
- 3 Probabilistic Uncertainty
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Why Uncertainty Matters

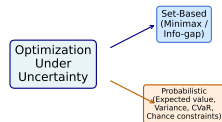
Core Problem

Previous chapters assumed the objective function $f(\mathbf{x})$ is *deterministic*.

In practice, uncertainty arises from:

- **Model error** — imperfect mathematical model
- **Manufacturing tolerances** — built design deviates from nominal
- **Environmental variation** — operating conditions change
- **Measurement noise** — noisy sensor readings
- **Distributional shift** — unknown inputs at test time

Chapter 20: Taxonomy of Uncertainty Approaches



Two main paradigms: set-based and probabilistic

Uncertain Objective Function

General Formulation

The objective depends on a **design vector** $\mathbf{x} \in \mathbb{R}^n$ and an **uncertainty vector** $\mathbf{z} \in \mathbb{R}^m$:

$$\text{minimize}_{\mathbf{x}} \quad f(\mathbf{x}, \mathbf{z})$$

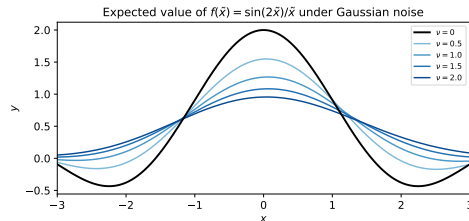
Uncertainty $\mathbf{z}$ cannot be controlled but may have known properties.

Running example (p. 428):

We evaluate $f(\mathbf{x}, \mathbf{z}) = f(\mathbf{x} + \mathbf{z}) = f(\tilde{\mathbf{x}})$

where $\tilde{\mathbf{x}} = \mathbf{x} + \mathbf{z}$.

- $\mathbf{z}$ may come from a *Gaussian distribution* $\mathcal{N}(0, \nu)$
- Expected value: $\mathbb{E}_{\mathbf{z}}[f(\mathbf{x} + \mathbf{z})] \neq f(\mathbf{x})$ in general
- Variance of $\mathbf{z}$ changes optimal design location



Objective $f(\tilde{\mathbf{x}}) = \sin(2\tilde{\mathbf{x}})/\tilde{\mathbf{x}}$ under varying noise

Two Paradigms for Handling Uncertainty

Set-Based Uncertainty

z belongs to a *known set* $\mathcal{Z}$:

$$z \in \mathcal{Z}$$

No probability distribution assumed.

Approaches:

- **Minimax** — optimize worst case
- **Info-gap** — maximize robustness radius

Natural when: physical bounds are known but statistics are not.

Probabilistic Uncertainty

z is drawn from a distribution P :

$$z \sim P$$

Probability distribution is modeled.

Approaches:

- **Expected value** minimization
- **Variance** / mean-variance trade-off
- **CVaR** — conditional value at risk
- **Chance constraints** — probabilistic feasibility

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Set-Based Uncertainty — Setup

Assumption

Set-based uncertainty assumes $\mathbf{z}$ belongs to a *set* $\mathcal{Z} \subset \mathbb{R}^m$. No probability distribution is assumed. One way to define $\mathcal{Z}$: use **interval constraints** on each component:

$$z_i \in [a_i, b_i] \quad \text{for all } i$$

Alternative: Inequality Constraints

$\mathcal{Z}$ can also be defined via a set of inequalities:

$$\mathcal{Z} = \{\mathbf{z} : g_j(\mathbf{z}) \leq 0, j = 1, \dots, m\}$$

e.g., ellipsoidal uncertainty sets.

Goal: find a design that performs well **for any** $\mathbf{z} \in \mathcal{Z}$.

Minimax Formulation

The **minimax** approach minimizes the worst-case objective:

$$\underset{\mathbf{x}}{\text{minimize}} \ f_{\text{mod}}(\mathbf{x}) \quad \text{where} \quad f_{\text{mod}}(\mathbf{x}) = \underset{\mathbf{z} \in \mathcal{Z}}{\text{maximize}} \ f(\mathbf{x}, \mathbf{z})$$

$f_{\text{mod}}(\mathbf{x})$ modified objective (worst-case value)
 $\mathcal{Z}$ uncertainty set

- Equivalent to a **nested** min-over-max optimization
- If f and $\mathcal{Z}$ are convex, the outer problem is convex
- With feasibility constraints: restrict to designs feasible *for all* $\mathbf{z} \in \mathcal{Z}$

Minimax Example (Example 20.1)

Setup

$$f(x, z) = f(\tilde{x}), \quad \tilde{x} = x + z$$

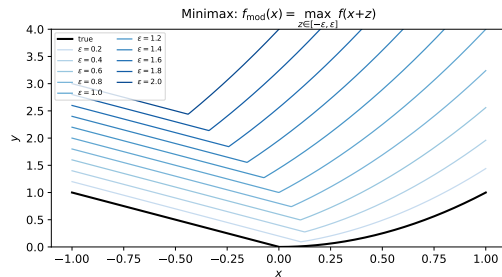
$$f(\tilde{x}) = \begin{cases} -\tilde{x} & \tilde{x} \leq 0 \\ \tilde{x}^2 & \text{otherwise} \end{cases}$$

Uncertainty set: $z \in [-\epsilon, \epsilon]$

Minimax objective:

$$f_{\text{mod}}(x) = \max_{z \in [-\epsilon, \epsilon]} f(x + z)$$

- $\epsilon = 0$: minimizer at $x^* = 0$
- As ϵ increases, minimizer shifts
- Robust minimizer $\neq$ noise-free minimizer



$f_{\text{mod}}(x)$ for $\epsilon = 0, 0.2, \dots, 2.0$

Minimax with Feasibility Constraints

Robust Feasibility

In problems with constraints $(\mathbf{x}, \mathbf{z}) \in \mathcal{F}$, a minimax approach considers only designs **feasible for all** $\mathbf{z} \in \mathcal{Z}$:

$$\underset{\mathbf{x}}{\text{minimize}} \ f_{\text{mod}}(\mathbf{x}) \quad \text{s.t.} \quad (\mathbf{x}, \mathbf{z}) \in \mathcal{F} \ \forall \mathbf{z} \in \mathcal{Z}$$

Equivalently, the robust feasible set is the **intersection**:

$$\mathcal{F}_{\text{robust}} = \bigcap_{\mathbf{z} \in \mathcal{Z}} \mathcal{F}_{\mathbf{z}}$$

Example 20.2 — Rotated Ellipse:

$(\mathbf{x}, \mathbf{z}) \in \mathcal{F}$ iff $z \in [0, \pi/2]$ and

$$(x_1 \cos z + x_2 \sin z)^2 + \frac{(x_1 \sin z - x_2 \cos z)^2}{16} \leq 1$$

Rotated ellipse feasible sets
(blue=union, red=intersection/robust)



20.2.2 Information-Gap Decision Theory

Alternative to Minimax

Instead of fixing $\mathcal{Z}$ and minimizing f_{mod} , **info-gap decision theory** asks: *how large can $\mathcal{Z}$ be while still achieving an acceptable objective value?*

Given a **performance requirement** f_c , find:

$$\hat{\epsilon}(\mathbf{x}) = \underset{\epsilon \geq 0}{\text{maximize}} \left\{ \epsilon : \max_{\mathbf{z} \in \mathcal{Z}(\epsilon)} f(\mathbf{x}, \mathbf{z}) \leq f_c \right\}$$

- $\hat{\epsilon}(\mathbf{x})$ robustness radius of design $\mathbf{x}$
- $\mathcal{Z}(\epsilon)$ uncertainty set of size ϵ
- f_c critical (acceptable) objective value

Key idea: Maximize the *robustness horizon* — the largest perturbation tolerable while maintaining acceptable performance.

Info-Gap: Example 20.3

Problem

$$f(x, z) = \tilde{x}^2 + 6e^{-\tilde{x}^2}, \quad \tilde{x} = x + z$$
$$\tilde{x} \in [-2, 2], \quad \mathcal{Z}(\epsilon) = [-\epsilon, \epsilon]$$

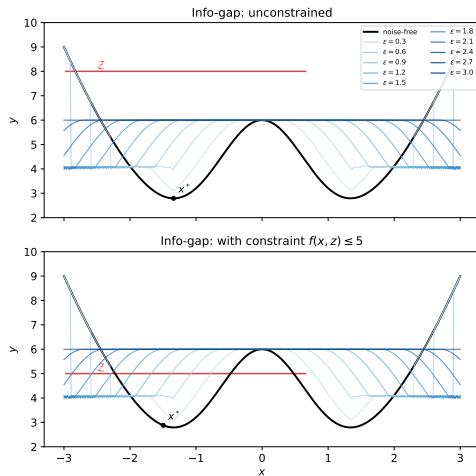
Top panel: Unconstrained info-gap.

Pure info-gap centers on a *suboptimal* region — risk aversion is excessive.

Bottom panel: With constraint $f(x, z) \leq 5$.

Adding a performance bound moves x^* to a region with *better* noise-free performance.

Blue lines = worst-case objective value for each ϵ .



Python: Minimax Optimization

```
1 import numpy as np
2 from scipy.optimize import minimize, differential_evolution
3
4 def f_tilde(xtilde):
5     """Piecewise objective f(x+z)."""
6     return np.where(xtilde <= 0, -xtilde, xtilde**2)
7
8 def f_mod(x, eps, n_z=200):
9     """Worst-case objective over z in [-eps, eps]."""
10    z_vals = np.linspace(-eps, eps, n_z)
11    return np.max(f_tilde(x + z_vals))
12
13 # Minimax: minimize worst-case over x in [-1, 1]
14 eps = 0.5
15 result = minimize(
16     lambda x: f_mod(x[0], eps),
17     x0=[0.0],
18     method='L-BFGS-B',
19     bounds=[(-1, 1)]
20 )
21 x_robust = result.x[0]
22 print(f"Robust minimizer (eps={eps}): x* = {x_robust:.4f}")
23 # eps=0.5 -> x* shifts right from 0 toward positive values
```

Minimax requires evaluating f over the full uncertainty set $\mathcal{Z}$ at each x — computationally more expensive than noise-free optimization.

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20.3 Probabilistic Uncertainty — Overview

Setup

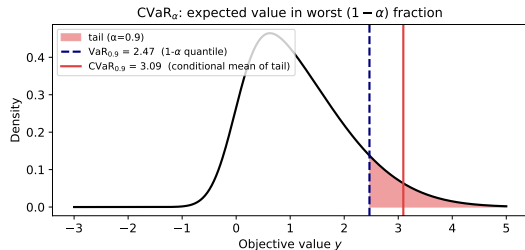
Assume $\mathbf{z}$ is drawn from a **probability distribution** P :

$$\mathbf{z} \sim P(\mathbf{z})$$

The objective $f(\mathbf{x}, \mathbf{z})$ is now a *random variable* $y = f(\mathbf{x}, \mathbf{z})$.

We optimize over the distribution of y using various risk measures:

- 1 **Expected value:** $\mathbb{E}_{\mathbf{z}}[f(\mathbf{x}, \mathbf{z})]$
- 2 **Variance:** $\text{Var}_{\mathbf{z}}[f(\mathbf{x}, \mathbf{z})]$
- 3 **Mean-variance trade-off:** combined
- 4 **Value at Risk (VaR):** quantile
- 5 **Conditional Value at Risk (CVaR):** tail mean
- 6 **Chance constraints:** $\Pr[g(\mathbf{x}, \mathbf{z}) \leq 0] \geq 1 - \delta$



20.3.1 Expected Value

Expected Value Objective

$$\underset{\mathbf{x}}{\text{minimize}} \mathbb{E}_{\mathbf{z} \sim P} [f(\mathbf{x}, \mathbf{z})] = \underset{\mathbf{x}}{\text{minimize}} \int f(\mathbf{x}, \mathbf{z}) P(\mathbf{z}) d\mathbf{z}$$

$P(\mathbf{z})$ probability density / distribution of uncertainty

$f(\mathbf{x}, \mathbf{z})$ objective under uncertainty realization $\mathbf{z}$

Key properties:

- If $f(\mathbf{x}, \mathbf{z}) = f(\mathbf{x}) + \mathbf{z}$ (additive noise), then $\mathbb{E}[f(\mathbf{x}, \mathbf{z})] = f(\mathbf{x}) + \mathbb{E}[\mathbf{z}]$ — noise shifts value, not optimum
- If $f(\mathbf{x}, \mathbf{z}) = f(\mathbf{x} + \mathbf{z}) = f(\tilde{\mathbf{x}})$, the expected value *smooths* the landscape (regularization effect)
- Zero-mean Gaussian output noise: $\mathbb{E}_{\mathbf{z} \sim \mathcal{N}(0, \Sigma)} [f(\mathbf{x}) + \mathbf{z}] = f(\mathbf{x})$ — *no effect*
- Zero-mean Gaussian input noise *does* shift optima

Expected Value: Gaussian Input Noise (Example 20.4)

Setup

Minimize $\mathbb{E}[f(\tilde{x})]$ where $f(\tilde{x}) = \sin(2\tilde{x})/\tilde{x}$
with $\tilde{x} = x + z$, $z \sim \mathcal{N}(0, \nu)$

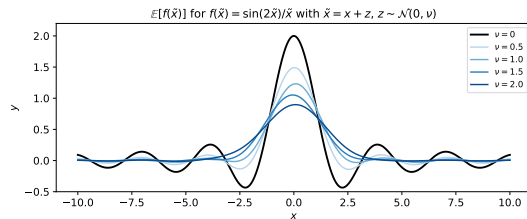
Zero-mean output noise: $\mathbb{E}[f(\mathbf{x}) + \mathbf{z}] = f(\mathbf{x})$
(no effect on objective)

Zero-mean *input* noise: *affected by ν*

$$\mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \Sigma)}[f(\mathbf{x}) + \mathbf{z}] = f(\mathbf{x})$$

$$\mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \Sigma)}[f(\mathbf{x} + \mathbf{z})] \neq f(\mathbf{x})$$

Increasing ν smooths the landscape:
local optima are averaged out.



Higher $\nu \Rightarrow$ smoother, shifted optima

20.3.2 Variance

Variance as a Risk Measure

Variance quantifies **spread** of the objective around its mean:

$$\text{Var}_{\mathbf{z}}[f(\mathbf{x}, \mathbf{z})] = \mathbb{E}[(f(\mathbf{x}, \mathbf{z}) - \mathbb{E}[f(\mathbf{x}, \mathbf{z})])^2]$$

Minimizing variance alone tends to over-penalize large deviations; often **combined with the mean**:

$$\underset{\mathbf{x}}{\text{minimize}} \quad \alpha \mathbb{E}[f(\mathbf{x}, \mathbf{z})] + (1 - \alpha) \sqrt{\text{Var}[f(\mathbf{x}, \mathbf{z})]} \quad \alpha \in [0, 1]$$

- $\alpha = 1$ pure expected-value optimization
- $\alpha = 0$ pure variance minimization
- $0 < \alpha < 1$ trade-off between mean and spread

Variance Example (Example 20.5)

Problem

$$f(x, z) = x^2 + z, \quad z \sim \text{Gamma}\left(\frac{2}{1+|x|}, 2\right)$$

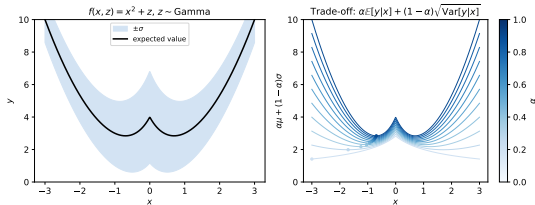
$$\text{Mean: } \mathbb{E}[z \mid x] = \frac{4}{1+|x|}$$

$$\text{Variance: } \text{Var}[z \mid x] = \frac{8}{(1+|x|)^2}$$

$$\text{Expected objective: } \mathbb{E}[f] = x^2 + \frac{4}{1+|x|}$$

Minima (EV only) at $x \approx \pm 0.695$

Adding variance penalty shifts minima *away* from ± 0.695 .



Left: $\pm\sigma$ band; Right: α -trade-off

Markowitz Portfolio Optimization (Example 20.6)

Mean-Variance in Finance

Portfolio $\mathbf{x}$ invests x_i in stock i .

Returns $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, portfolio return $y = \mathbf{x}^\top \mathbf{z}$.

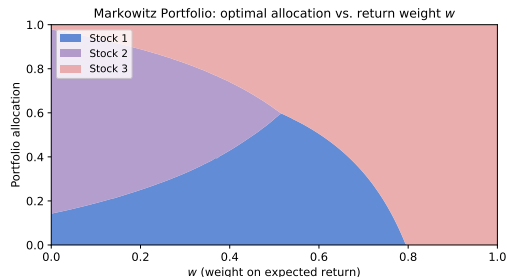
Markowitz problem:

$$\underset{\mathbf{x}}{\text{minimize}} \quad -w\mathbf{x}^\top \boldsymbol{\mu} + (1-w)\mathbf{x}^\top \boldsymbol{\Sigma} \mathbf{x}$$

$$\text{s.t.} \quad \mathbf{x}^\top \mathbf{1} = b, \mathbf{x} \geq \mathbf{0}$$

$\boldsymbol{\mu} = [0.26, 0.08, 0.74]^\top$ expected returns
 $w \in [0, 1]$ weight on return vs. risk

This is a **quadratic program** (Chapter 13).



As $w \rightarrow 1$: high-return stock 3 dominates.

As $w \rightarrow 0$: low-variance stock 2 dominates.

Python: Mean-Variance Portfolio Optimization

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 mu      = np.array([0.26, 0.08, 0.74])
5 Sigma = np.array([[0.21, 0.03, 0.01],
6                   [0.03, 0.06, 0.04],
7                   [0.01, 0.04, 0.94]])
8 b = 1.0  # budget
9
10 def markowitz(w):
11     """Optimal Markowitz portfolio for return weight w."""
12     def obj(x): return -w * x @ mu + (1-w) * x @ Sigma @ x
13     def jac(x): return -w * mu + 2*(1-w) * Sigma @ x
14     res = minimize(obj, np.ones(3)/3, jac=jac,
15                   method='SLSQP',
16                   constraints=[{'type': 'eq',
17                                'fun': lambda x: x.sum() - b}],
18                   bounds=[(0, None)]*3)
19     return res.x
20
21 # w=1.0: maximize expected return
22 x_ret  = markowitz(1.0)
23 # w=0.0: minimize variance
24 x_safe = markowitz(0.0)
25 print(f"Max-return portfolio: {x_ret.round(3)}")
26 print(f"Min-variance portfolio: {x_safe.round(3)}")
```

20.3.3 Value at Risk (VaR)

Definition

The **Value at Risk** at level α is the α -quantile of the objective value distribution $y = f(\mathbf{x}, \mathbf{z})$:

$$\text{VaR}_\alpha(\mathbf{x}) = \inf\{y : \Pr[f(\mathbf{x}, \mathbf{z}) \leq y] \geq \alpha\}$$

$\alpha \in (0, 1)$ confidence level (e.g. 0.95)
 VaR_α objective value exceeded with probability $1 - \alpha$

Numerical example: $\alpha = 0.95$, samples of $y = f(\mathbf{x}, \mathbf{z})$:

- $\text{VaR}_{0.95}$ is the 95th percentile of y
- Only 5% of outcomes exceed $\text{VaR}_{0.95}$

Limitation: VaR is not coherent — it ignores severity of losses *beyond* the quantile threshold.

20.3.4 Conditional Value at Risk (CVaR)

Definition

CVaR_α (also called *expected shortfall* or *superquantile*) is the **expected value of y in the worst $(1 - \alpha)$ fraction**:

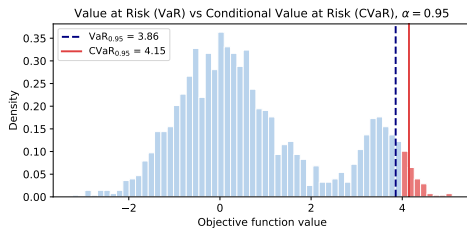
$$\text{CVaR}_\alpha(\mathbf{x}) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(\mathbf{x}) du$$

Equivalently:

$$\text{CVaR}_\alpha(\mathbf{x}) = \mathbb{E}[f(\mathbf{x}, \mathbf{z}) \mid f(\mathbf{x}, \mathbf{z}) \geq \text{VaR}_\alpha(\mathbf{x})]$$

Advantages over VaR:

- Convex and sub-additive (coherent risk measure)
- Less sensitive to estimation errors
- Captures severity of extreme outcomes



Red region = worst $(1 - \alpha)$ fraction.

CVaR = mean of red region.

CVaR: Tractable Reformulation

Rockafellar-Uryasev Formula (2000)

CVaR admits a tractable optimization formulation:

$$\text{CVaR}_\alpha(\mathbf{x}) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{1-\alpha} \mathbb{E}_{\mathbf{z}} [\max(f(\mathbf{x}, \mathbf{z}) - \tau, 0)] \right\}$$

τ auxiliary variable (optimal $\tau^* = \text{VaR}_\alpha$)

Key properties:

- Always $\text{CVaR}_\alpha \geq \text{VaR}_\alpha \geq \mathbb{E}[f]$
- CVaR is a **coherent risk measure**
- Monte Carlo estimation: sort samples, take mean of top $(1 - \alpha)$

Numerical Example

If f values (sorted) are $[1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 6.0, 8.0, 9.0, 10.0]$ with $\alpha = 0.90$:

$$\text{VaR}_{0.90} = 9.0, \quad \text{CVaR}_{0.90} = \frac{9.0 + 10.0}{2} = 9.5$$

Python: CVaR Estimation and Optimization

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 def cvar_mc(x, f_func, dist_sample, alpha=0.95, n=1000):
5     """Monte Carlo CVaR estimate."""
6     z_samples = dist_sample(n)           # shape (n,)
7     y = np.array([f_func(x, z) for z in z_samples])
8     y_sorted = np.sort(y)
9     k = int(np.ceil(alpha * n))           # index of VaR
10    var_alpha = y_sorted[k]
11    cvar_alpha = y_sorted[k:].mean()       # CVaR = mean of tail
12    return cvar_alpha, var_alpha
13
14 # Example:  $f(x, z) = (x - z)^2$ ,  $z \sim N(0, 1)$ 
15 f_func = lambda x, z: (x - z)**2
16 dist_sample = lambda n: np.random.randn(n)
17
18 # Minimize CVaR_0.95 over x in [-3, 3]
19 def neg_cvar(xvec):
20     return cvar_mc(xvec[0], f_func, dist_sample, alpha=0.95)[0]
21
22 res = minimize(neg_cvar, [0.0], method='Nelder-Mead',
23               options={'xatol':1e-4, 'fatol':1e-4})
24 print(f"CVaR-optimal x = {res.x[0]:.4f}")
25 # Expect x* near 0 (minimizing expected squared deviation from  $z \sim N(0, 1)$ )
```

20.3.5 Statistical Feasibility (Chance Constraints)

Chance Constraint Formulation

Instead of requiring $g(\mathbf{x}, \mathbf{z}) \leq 0$ for *all* $\mathbf{z}$, require feasibility with **high probability**:

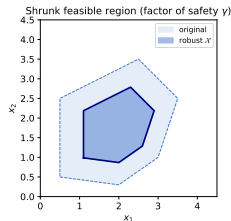
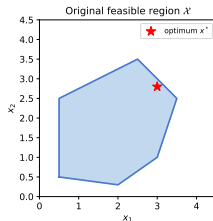
$$\Pr_{\mathbf{z} \sim P}[g(\mathbf{x}, \mathbf{z}) \leq 0] \geq 1 - \delta$$

$\delta \in (0, 1)$ acceptable violation probability

$1 - \delta$ required confidence level

Approaches:

- **Analytical**: closed-form if $\mathbf{z}$ Gaussian and g linear
- **Sample average approximation**: replace $\Pr[\cdot]$ with fraction of satisfied samples
- **Factor of safety**: scale constraints $g(\mathbf{x}) \leq g_{\max}/\gamma$ with $\gamma > 1$



Shrinking feasible region via factor of safety γ

Statistical Feasibility: Factor of Safety

Factor of Safety

Rewrite constraint $g(\mathbf{x}) \leq g_{\max}$ as:

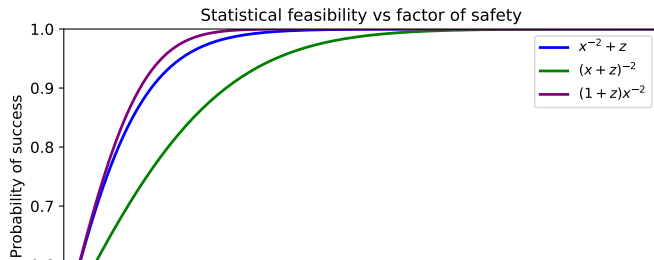
$$\gamma g(\mathbf{x}) \leq g_{\max}, \quad \gamma > 1$$

where γ is the **factor of safety**.

This ensures the design stays below $g_{\max}/\gamma < g_{\max}$, providing a safety margin.

Beam design example (Exercise 20.2):

Minimize cross-section $f(x) = x^2$; stress $g(x) = x^{-2} \leq 1$; optimal $x^* = \sqrt{\gamma}$.



The Six-Sigma Standard (Exercise 20.3)

Definition

Six-sigma requires that feasibility violations occur only when $|z| > 6\sigma$ (standard deviations):

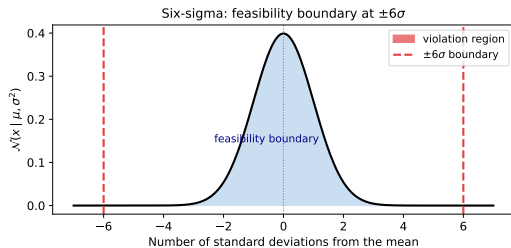
$$\Pr[g(\mathbf{x}, \mathbf{z}) \leq 0] \geq 1 - \Pr[|z| > 6] \approx 1 - 1.973 \times 10^{-7}\%$$

Extremely stringent: only $\approx 2 \times 10^{-9}$ of outcomes may violate.

Six-sigma exercise: Find x^* such that (x, z) feasible for all $|z| \leq 6$:

$$\text{minimize } x_1 \quad \text{s.t. } e^{x_1} \leq x_2 + z \leq 2e^{x_1}$$

Solution: $x_1 = \ln 12 \approx 2.485$, $x_2 = 18$.



Only tails beyond 6σ are infeasible (red)

Python: Chance Constraints via Sample Average Approximation

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 np.random.seed(42)
5 N = 500 # number of scenarios
6
7 # Sample  $z \sim N(0, 0.01)$  ( $\sigma = 0.1$ )
8 z_samples = np.random.randn(N) * 0.1
9
10 def chance_constraint_violation(x, delta=0.05):
11     """
12     Returns fraction of scenarios that violate  $g(x,z) \leq 1$ .
13     Constraint:  $x^{-2} + z \leq 1$ 
14     """
15     g_vals = x**(-2) + z_samples
16     violation_rate = np.mean(g_vals > 1.0)
17     return violation_rate
18
19 # Minimize  $f(x) = x^2$  subject to:  $Pr[x^{-2} + z \leq 1] \geq 0.95$ 
20 # i.e.,  $violation\_rate \leq 0.05$ 
21 def objective(xvec): return xvec[0]**2
22 def constraint(xvec):
23     return 0.05 - chance_constraint_violation(xvec[0])
24
25 res = minimize(objective, [1.5],
26               constraints=[{'type': 'ineq', 'fun': constraint}],
27               bounds=[(0.5, 5.0)], method='SLSQP')
```

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Chapter 20 Summary I

Key Concepts

Uncertain objective $f(\mathbf{x}, \mathbf{z})$ depends on uncontrollable $\mathbf{z}$

Set-based $\mathbf{z} \in \mathcal{Z}$ — no distribution assumed

Probabilistic $\mathbf{z} \sim P$ — distribution is modeled

Set-Based Methods:

- **Minimax:** $\text{minimize}_{\mathbf{x}} \max_{\mathbf{z} \in \mathcal{Z}} f(\mathbf{x}, \mathbf{z})$
Optimizes worst-case performance
- **Robust feasibility:** design feasible for *all* $\mathbf{z} \in \mathcal{Z}$
Robust set = intersection of feasible sets
- **Info-gap:** maximize robustness radius $\hat{e}(\mathbf{x})$
Requires a performance threshold f_c

Chapter 20 Summary II

Probabilistic Methods:

- **Expected value:** $\text{minimize}_{\mathbf{x}} \mathbb{E}_{\mathbf{z}}[f(\mathbf{x}, \mathbf{z})]$
Smooths landscape; shifts optima under input noise
- **Variance:** $\text{minimize}_{\mathbf{x}} \text{Var}_{\mathbf{z}}[f(\mathbf{x}, \mathbf{z})]$
Risk-neutral; combined with mean in practice
- **Mean-variance trade-off:** $\alpha \mathbb{E}[f] + (1 - \alpha) \sqrt{\text{Var}[f]}$
Markowitz portfolio is the canonical example
- **VaR:** α -quantile of objective
Easy to understand; not coherent
- **CVaR:** expected value in worst $(1 - \alpha)$ tail
Coherent; tractable via Rockafellar-Uryasev formula
- **Chance constraints:** $\Pr[g(\mathbf{x}, \mathbf{z}) \leq 0] \geq 1 - \delta$
Statistical feasibility; factor of safety approach

Risk Measures Comparison I

Method	Distribution?	Tail-sensitive?	Convex?
Minimax	No	Yes (worst-case)	If $f, \mathcal{Z}$ convex
Info-gap	No	No (avg-case)	Generally no
Expected value	Yes	No	If f convex
Variance	Yes	Partially	If f convex
VaR	Yes	No (threshold)	No
CVaR	Yes	Yes	Yes
Chance constr.	Yes	No (probability)	Special cases

Choosing a Method

- **Physical safety bound known** $\Rightarrow$ minimax or chance constraint
- **Distribution well-characterized** $\Rightarrow$ expected value or CVaR
- **Finance / risk management** $\Rightarrow$ CVaR (coherent, convex)
- **Engineering reliability** $\Rightarrow$ factor of safety or six-sigma
- **Trade-off desired** $\Rightarrow$ mean-variance (Markowitz)

Exercises

Exercise 20.1

Consider the objective shown below with three design points a , b , c . The objective is to minimize $\mathbb{E}_z[f(x+z)] - \sqrt{\text{Var}_z[f(x+z)]}$.

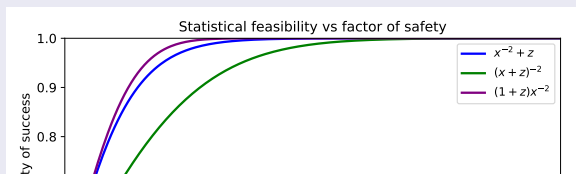
The optimal design is $x^* = a$ (minimizes both mean and std).

Exercise 20.2: Factor of Safety — Beam Design

Minimize $f(x) = x^2$ (cross section area) subject to stress $g(x) \leq \sigma_{\max} = 1$.

Optimal: $x^* = \sqrt{\gamma}$. Three uncertainty models:

- ① $g(x, z) = x^{-2} + z$ (stress uncertainty)
- ② $g(x, z) = (x + z)^{-2}$ (construction tolerance)
- ③ $g(x, z) = (1 + z)x^{-2}$ (material property)



Exercise 20.3: Six-Sigma Feasibility

Problem

Minimize x_1 subject to:

$$e^{x_1} \leq x_2 + z \leq 2e^{x_1}, \quad z \sim \mathcal{N}(0, 1)$$

Find x^* such that (x, z) is feasible for all $|z| \leq 6$.

Solution:

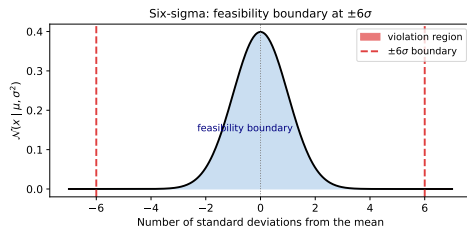
x_2 must satisfy $x_2 - z \geq e^{x_1}$ and $x_2 + z \leq 2e^{x_1}$ for all $|z| \leq 6$.

Robust choice: $x_2 = 1.5e^{x_1}$ (center of $[e^{x_1}, 2e^{x_1}]$).

Width of feasible window: e^{x_1} ; must accommodate 12σ :

$$e^{x_1} \geq 12 \Rightarrow x_1 = \ln 12 \approx 2.485$$

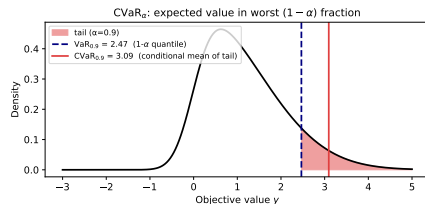
$$x_2 = 1.5e^{x_1} = 18$$



Conclusions

Key Takeaways

- Uncertainty is *inherent* in real-world optimization
- **Set-based** methods (minimax, info-gap) require no probability model — robust but often conservative
- **Probabilistic** methods use distributional information — more efficient when distribution is known
- **CVaR** is preferred over VaR: coherent, convex, tractable
- **Chance constraints** directly specify reliability requirements
- Methods can be combined (e.g., CVaR with chance constraints)
- Monte Carlo sampling enables optimization for any distribution



CVaR captures what VaR misses:
severity of extreme outcomes.